

Securing data of biotechnological laboratories using blockchain technology

Krzysztof Misztal^{1,2}[0000–0003–2480–3505],
Tomasz Szulalec^{1,2}[0000–0001–6217–4274], and Aleksandra
Kubica-Misztal²[0000–0001–5264–990X]

¹ Jagiellonian University, Poland
{krzysztof.misztal,t.szulalec}@uj.edu.pl
² diCELLa Ltd, Poland
a.kubica-misztal@dicella.com

Abstract. A few years ago blockchain technology was used in cryptocurrency. Nowadays, a variety of diverse areas are seeing the benefits of applying this technological approach to their needs. One way transactions without reverse mode is making blockchain a desirable platform for maintaining data. The authenticity, transparency and authorization of it make it ideal for healthcare or laboratory data systems. Research data should be authorized by a specific employee and never changed. The information collected should be never forged or falsified. Designing a blockchain system with access and permission rules is ideal in such a situation. In this article, we present the adaptation of blockchain technology to medical research laboratories and diagnostics. Afterwards, an RSA signature user can store information of any type and size. The new SHA-3 hash function is used to bind blocks together. This technological path makes laboratory workflow more efficient and fulfills restriction on medical laws.

Keywords: blockchain · laboratories · database

1 Blockchain system

Blockchain system. In 2008, Satoshi Nakamoto published a paper [5] describing bitcoin, which is now referred to as the first cryptocurrency in the world. The prefix *crypto* is derived from the word cryptography. The bitcoin is a complete and complex system which provides the ability to perform anonymous and cashless transactions over the Internet. Bitcoin is based on a decentralized settlement book, which can be managed by any of the system's users. Such users are called miners because their reward for the management is the newly mined currency. The miners gather information about transactions and validate whether they can be performed. Then they choose the transactions to store in the settlement book. This book is known as a blockchain. Thanks to the used cryptography algorithms, it is impossible to make any transaction as another user. Moreover, if a transaction is stored in the blockchain, no one can change or delete it.

It is worth noticing that while all bitcoin transactions are stored in the blockchain unencrypted, it is possible to keep encrypted data in the blockchain.

1.1 Bitcoin technology

There are several implementations of blockchain and we will focus on the well known precursor bitcoin. For the more information the reader is referred to [2]. The sellers or payers that want to make a transaction must submit their action to blockchain. Everyone that is connected to the blockchain system receive the information about the transaction. Receivers are called miners and are the clients which validate the action by algorithm. The validation by cryptographic algorithm is compound with two computation steps. Miners provide the computation power. Software for mining is free and simple. Moreover no certification is required so everyone can volunteers with their computers. Miner must validate existence of the bitcoin and that can be used for transaction. Then he checks the form and accepts the transaction. If the miner validate the whole block he is rewarded with a bitcoin. Block is has the detailed information about validation processes of transactions. The block is stored in the blockchain after it is validated by a fixed number of miners. What is more within every block are timestamps and a mathematically generated complex variable sums called hashes that secure the interruption in data block. These cryptographic hashes of blocks are compounded of preceding and current block. This distinct and unique sum make up the security signature that combine blocks into blockchain. The irreversibility of hash code makes blockchain immutable. This way we obtain digital trust. Decentralized validation of actions and saving them to the history makes mediator entity like banks useless to perform transaction. Example:

1. John wishes to pay bitcoins to Alice.
2. John's transaction is made.
3. Transactions are combined into a new block.
4. Blocks are broadcasted to miners.
5. Miners validates and accepts the transactions.
6. New block is encrypted with hash of last block.
7. New block become a part of blockchain.
8. Alice receives bitcoins.

1.2 Sensitive data in blockchain

Decentralized system can be widely used. The greatest benefit is that there is no need of third parties. As we have seen before, the decentralized data may be secured and immutable. Nowadays, most organizations collect our personal data in order to optimize their services. They want to predict our needs and create our digital personal image thus personal information in every aspect is a valuable resource in today economy. Although there are many benefits to a personalized system, the concern for our privacy is doubtful. What is more, now more often organizations have little control over the personal and sensitive

data. That is why these information should not be handled by third-parties. The centralized system is vulnerable to hacker attacks, as well as phishing. The solution is for the user to own and be responsible for providing its information. We can do this by applying blockchain storage solution proposed by [8]. Authors proposed personal data management system that a advantage of blockchain and off-blockchain storage. This requires several key aspects: data ownership, data transparency audibility and fine-grained access control. The proposed platform devided to users that own and control their personal information and the guest services. Every user has information about collocated data and how is it accessed by organization. Services have delegated premissions from users to access their data. What is more, the owner of the data is able to make changes to the set of permissions to deny the access to data. This solution is similar to that of mobile apps. It will not have an impact on the user-interface because the access-control policies are stored in a blockchain. Only users are able to make changes in the blockchain.

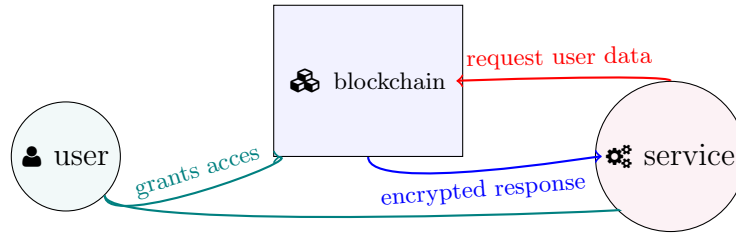


Fig. 1. Simplified schema of the above grant-access data blockchain.

Some ideas about blockchain application in health care can be seen in [2].

2 Laboratory

The results obtained in GMP and GLP laboratories have to be reliable. Any modification of partial results can make the final result fail. It is the reason why the lab records have to be collected precisely and without any forgery. It is also important to remember every the author of every record. The authorship has to be beyond any doubt, which means that nobody can create a record on behalf of another user and no user can deny his own authorship of any of his records. Moreover, the documentation has to be immutable so any stored data cannot be modified nor deleted. These requirements make users keep exact records, which objectively increases the reliability of the laboratory.

Solving this problem required us to cooperate with the employees of biotechnological laboratories and diagnostics. It required a great amount of time to translate the real life problem to algorithmic technological approach.

There were several key aspects for lab employees:

- constant access to warehouse- with restriction and access rules,
- rules of processing the production - to improve the process,
- rules of gathering lab results - the authorship of obtained data,
- rules of reporting everyday work - to avoid forgery and falsification,
- communication with clients.

All these aspects have a significant impact on the lab workflow.

2.1 Blockchain application

The author of bitcoin, [5], stated that there can be a central authority which produces coins and validates the transaction. It brings up the problem that the authority would have infinite knowledge regarding every transaction. This problem can be solved in a very tricky way. Before we attempt to explain the solution to this let us focus on transaction that we define in blockchain for GMP and GLP laboratories. We want every data information to be unmodified and has its author. In our platform the transaction of a single information would represent a coin in bitcoin system.

Let us consider a scenario in which user A has created a note about some topic. Our system needs to remember the authorship of that note as well as the fact that user A cannot deny any part of that note. That is why the user certifies the note by his digital signature. The different B user stated that the note has faulty so he want to make a correction. He can do this but not in the obvious way. User B takes the note, adds the calculated hash code from that note, add a correction to the note. After his digital signature he submits the information. In this way every user can validate which part is A's and which is B's. We see a compatibility between bitcoin, crypto-coin and our data note. The correction mechanism made by B is similar to case of spending same cryptpo-coin twice. We observe that system witch verifies folding signatures would solve this problem, the same as in bitcoin scenario.

A different aspect is when a need to create information arises. Coins can be made in a specific regularized way. The information stored in our platform should be for a every authorized user. That is why we need a second type of coin that would represent the permission access for users. That is why every authorized laboratory user would have the ability to add a digital key of his coworker allowing him to make transitions in the blockchain. As well as every authorized user it the same way can deny specific user to access the system. To sum up, in the bitcoin system the information about from who to whom the coin is sent is stored. In out platform we store the information about:

- users *digital keys* with permission information - we track if the user is allowed to signature the transaction,
- data information - we track authorship of the laboratory data; this solution is general that any type of lab data can be stored.

2.2 Technological details

Programming language and libraries. We choose specific cryptographic packages and programming language Python3 so we can create application on any technology. Although python apps are not as efficient as in C language but they are more portable. What is more professional programmers that are working with cyber-security chooses also python. In the prototype we used Django framework and cryptography python package because of the SHA-3 implemented hash function. It allows to simple and effective programming and is extending rapidly. After the theoretical analysis ensures us that these algorithms are sufficient to our needs.

Block hash function. The base and essential ingredient for the blockchain is a well-defined hash function. The National Institute of Standards and Technology (NIST) published two documents [1] and [3]. Where they certified only three hash functions:

- SHA-1,
- SHA-2,
- SHA-3.

The SHA-3 is technologically completely different. The SHA-1 and SHA-2 is based on the same mathematical mechanism. In 2017 Google announced [7] that SHA-1 is broken and should not be used for security. In comparison to SHA-1 and SHA-2, chosen SHA-3 is new and free of bugs algorithm.

User digital signature. Next important step in creating the blockchain is the algorithm for digital signatures and its validation. Here, the key solution is an asymmetric one-way cryptographic function. The National Institute of Standards and Technology (NIST) in [4] have defined three cryptographic algorithms that used digital signatures:

- RSA,
- DSA,
- ECDSA.

Due to the controversy with elliptic curves proposed by cryptographic authority Bruce Schneier [6] we chose the RSA algorithm with a key size of 4096 bits.

Blocks implementation details. To avoid the future problems we designed the blocks in the form of dictionary with a predefined set of keys. Those objects can be easily serialized and deserialized to the JSON format. Listing 1.1 shows seven keys for dictionary-blocks.

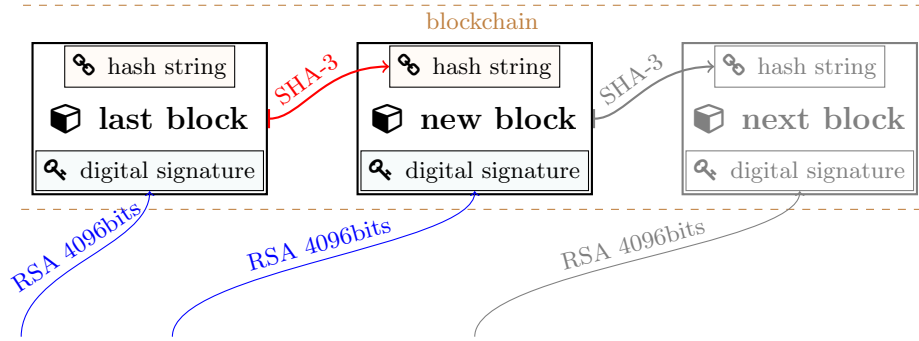


Fig. 2. Hash linking and RSA signature in blocks.

```
class BlockField(Enum):
    BLOCK_TYPE = 0
    TIMESTAMP = 1
    AUTHOR = 2
    SIGNATURE = 3
    OBJECT = 4
    PREVIOUS_HASH = 5
    NONCE = 6
```

Listing 1.1. Enum describing all parameters in block object.

We also defined five types of blocks shown at listing 1.2.

```
class BlockType(Enum):
    GENESIS = 0
    INSERT_BLOCK = 1
    UPDATE_BLOCK = 2
    INSERT_KEY = 3
    REVOKE_KEY = 4
```

Listing 1.2. Enum describing possible types of blocks in the chain.

The *genesis block* is a first one in the chain. There can be only one *genesis block* in the blockchain. It contains the system administrator key and it cannot be rejected. Public key is controlled by admins managing the software, not in the hands of laboratory employees.

2.3 Platform

We have ended the work of implementation of blockchain technology prototype, moreover, we have made an unconventional modification of blockchain technology in the way that is adapted to the objectives given by our theoretical analysis.

We validated time resources that were saved by our system. There was a significant improvement. System was effective to time-consuming and resource-consuming problems. The designed platform was verified in laboratory conditions. We observed that the designed solution had an impact on the laboratory workflow.

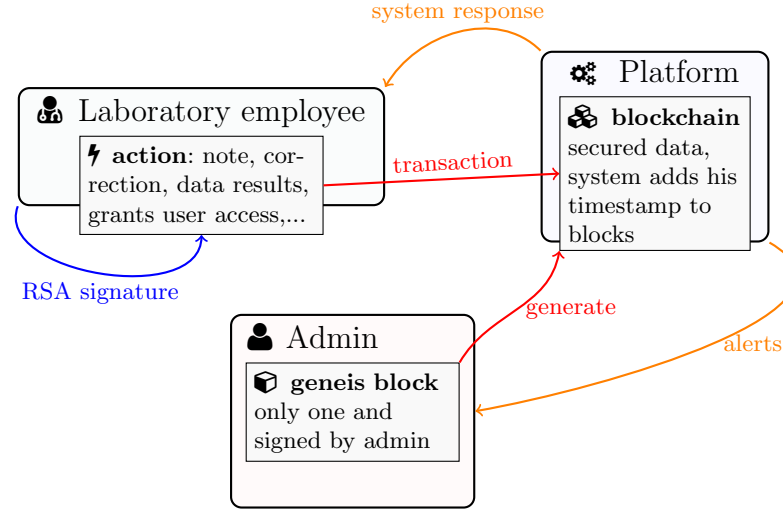


Fig. 3. Proposed laboratory blockchain platform usage. Platform is validating and storing signed and secured transaction. Additional lab-supporting algorithms perform analyses for the users basing on the blockchain data. Alerts are send when needed.

We have confirmed the effectiveness in every aspect of the use cases, as well as in the performance of our system. The processing time and memory usage by blockchain mechanism were not computation demanding.

Our designed template platform with small customization provides benefits that are widely desired in the laboratories. Below is the list of advantages that comes from proposed blockchain system.

1. Data are secured against unauthorized changes.
2. Blockchain security of data makes them authentic and repudiation.
3. Data stored in blockchain are transparent.
4. Due to GMP CFR 21 part 11 system allows to electronic documentation conducting.
5. System fulfills the acts of biobank laws.
6. System helps realization in tissue and cell procurement in order with doctor and dentist's law.
7. System fulfills the acts of pharmaceutic laws in terms of production advanced medicinal products.

8. System satisfies UE documents regarding production and medical therapy (transplanting cells, tissue and organs).
9. System additional algorithms archive the statistics improving the lab workflow.
10. System helps in stable managing the laboratory storage magazine.
11. System build in analytical algorithms save time for employees - list demands verification, production.

2.4 System stability

The author in [5] introduce the *timestamp server* idea which binds blocks chronologically. In our proposed system there is only one *timestamp server*. The web or physical access to the server is limited to minimum. System takes the created and edited data in laboratory as well as key of users whose can submit these data. *Timestamp server* will sort transactions in the chronological order. When the new data is added system is validating its propriety. For example:

- if the note was made before,
- if edited content of the note corresponds to the last one in blockchain,
- if a user is allowed to add a note - system verifies if user *digital key* is active and permitted to add a content.

We have made stability list to determine if the blockchain is corrupted. Below is the checklist that is validated by system when inserting a block to blockchain:

1. the first block is not a *genesis block*,
2. more then one block is *genesis block*,
3. *prev_hash* filed for *genesis block* is different then expected,
4. *timestamp* from name of *block* is not corresponding to the content of block,
5. *prev_hash* value of the *block* is not compatible with its *previous block*,
6. *hash string* do not begins with fixed length of zeros,
7. user *digital key* is added twice,
8. user *digital key* is deleted twice,
9. *genesis key* is deleted,
10. block added by unauthorized user - due to no *digital key*,
11. block added by unauthorized user - due to *digital key* rejection,
12. lack of mandatory filed in *block*.

We tried to hack our system. We used brute-force attacks but luckily without any success. The cryptographic packages the we have chosen in the carrying out of our system are dedicated to this kind of unwanted hacking access. Moreover the SHA-3 algorithm is used as the hash function is the one of the safest, most secure and the most efficient hash algorithms. In the case of non-quantum computers the SHA-3 algorithm should not be broken in the 20-30 years.

3 Conclusions

The blockchain technology is widely used in accounting and banking, but not exclusively in those fields. There are many projects adapting this technology in many different areas of daily lives. Our project, by using blockchain technology in the widely understood area of healthcare, is among them.

The requirements for the documentation in biotechnological labs cannot be enforced when records are being kept on paper. If one read the above descriptions carefully, one can notice that the requirements for lab records are satisfied by the main features of blockchain. In our talk we would like to present our innovative system for securing data of biotechnological laboratories using blockchain technology. Our presentation will show what data is stored in our blockchain and how the correctness of newly added *transactions* is checked.

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